

PayWise: Mindful UPI & Wealth Ecosystem

Comprehensive Project Documentation

Executive Summary

PayWise is a production-grade, PhonePe-inspired fintech application built with **Flutter 3 (Web & Mobile)** and powered by **Firebase** (Authentication & Cloud Firestore). Designed around the ethos of *mindful financial management*, PayWise blends rapid UPI payments, automated bill settlements, intelligent debt management, physical-backed digital wealth, personalized insurance coverage, instant micro-lending, and gamified cashback rewards into a coherent, ultra-clean mobile-first experience.

1. System Architecture & Tech Stack

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PRESENTATION LAYER	
Flutter 3.x (Web / iOS / Android)	GoRouter Navigation Shell
Responsive 430px Max-Width Center Container	Vanilla Custom Tokens

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STATE MANAGEMENT LAYER	
ChangeNotifier (AppProvider)	Centralized Reactive State Engine
Firestore Stream Subscriptions (Transactions, Portfolio, Profile, etc.)	

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DATA ACCESS LAYER	
FirestoreService (Singleton Gateway)	AuthService (Email & Demo Auth)
PricingEngine (Pure Dart Business Logic - Limits, Fees, Schedules)	

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BACKEND SERVICES

- Firestore Authentication (Email & Anonymous Guest Sign-In)
- Google Cloud Firestore (9 Real-Time Collections & Structured Documents)
- Firestore Hosting (Global CDN with Zero-Cache Web App Serving)

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Core Technology Stack:

- **Framework:** Flutter 3.24+ (Channel Stable)
- **Language:** Dart 3.5+
- **Navigation:** `go_router`: ^14.6.2 (StatefulShellRoute with 5 indexed branches)
- **State Management:** `provider`: ^6.1.2 (`ChangeNotifier` & reactive streams)
- **Database & Auth:** `firebase_core`: ^3.8.0, `cloud_firestore`: ^5.5.0, `firebase_auth`: ^5.3.3
- **Charts & Data Visualization:** `fl_chart`: ^0.69.2 (Interactive Donut spending breakdown)
- **Vector Graphics:** `flutter_svg`: ^2.0.16 (45 bespoke line icons, zero emojis)
- **Typography:** Google Fonts **Plus Jakarta Sans**
- **Testing:** `flutter_test` (23 automated unit and widget test cases)

2. Design System & Theme Tokens

PayWise adheres to an exact mathematical design token system defined in `lib/core/theme/`:

Token Category	Token Constant	Hex Value	Purpose & Usage
Primary Color	<code>AppColors.primary</code>	<code>#5B259F</code>	PhonePe signature deep royal purple
Primary Dark	<code>AppColors.primaryDark</code>	<code>#401375</code>	Hero gradient end, active tap states
Primary Light	<code>AppColors.primaryLight</code>	<code>#8C4FE6</code>	Highlights, active chips, icon backgrounds
Accent Green	<code>AppColors.accentGreen</code>	<code>#10B981</code>	Success states, credit inflows, savings badges
Alert Orange	<code>AppColors.alertOrange</code>	<code>#F59E0B</code>	PayPause mindful friction, cautionary alerts
Danger Red	<code>AppColors.dangerRed</code>	<code>#EF4444</code>	High budget exceedances, transaction failures
Background Neutral	<code>AppColors.bgNeutral</code>	<code>#F8F9FA</code>	Page canvas background
Surface Pure	<code>AppColors.surfaceWhite</code>	<code>#FFFFFF</code>	Card containers, bottom sheets, dialogues
Text Primary	<code>AppColors.textPrimary</code>	<code>#1E293B</code>	High-contrast body copy and headings
Text Secondary	<code>AppColors.textSecondary</code>	<code>#64748B</code>	Subtitles, helper text, timestamps
Border Subdued	<code>AppColors.borderSubdued</code>	<code>#E2E8F0</code>	Divider lines, outline stroke (1px)

3. Business Logic & Pricing Engine

All operational financial rules are strictly isolated in `lib/core/pricing_rules.dart`:

1. Daily Bank Transfer Cap:
- Limit: **■1,00,000 per rolling 24-hour day**.
 - Rule: Transfers above **■1,00,000** are rejected immediately with descriptive guidance.
2. Mindful PayPause Threshold:
- Soft Friction: **■2,000**.
 - Rule: Payments of **■2,000** or greater trigger an intentional pause bottom sheet displaying balance impact, recent category spending, and a 3-second reflection timer before confirmation.
3. Credit Card Bill Processing Fees:
- First 2 card bill settlements per calendar month: **■0.00 (Free)**.
 - Subsequent settlements (3rd card onward): **1.0% convenience fee**.
4. Loan Eligibility Multiplier:
- Formula: $\text{Approved Limit} = \text{\$}\text{Monthly Inflow} \times 2.5\text{\$}$.
 - Max Cap: **■2,50,000** at a competitive 11.5% APR.
5. Gold Investment Rate:
- Real-time ticking 24K 99.9% pure gold price simulated with $\pm 0.1\%$ market variance.
 - Live buy and sell spreads with instant locker updates.

4. Firestore Database Architecture (9 Collections)

Collection Name	Key Attributes	Description
users	name, phone, email, upiId, walletBalance, kycStatus	Core user identity & live wallet balance
transactions	amount, type, counterparty, category, timestamp, status, referenceId	Audit-trailed payment ledger entries
contacts	name, phone, upiId, avatarUrl, isFavorite, lastSentAmount	Frequent recipients and phone directory
bills	billerName, category, amountDue, dueDate, autopayEnabled, scheduleType	Utility, mobile, and broadband subscriptions
credit_cards	bankName, cardNumberMasked, totalDue, minDue, dueDate, freeBillsUsed	Linked credit cards and payment fee counters
portfolio	totalValue, dailyReturnRate, goldGrams, mutualFundsValue, stocksValue	Wealth holdings and asset distribution
insurance_policies	title, type, provider, coverageAmount, premiumAmount, status	Active health, vehicle, and life coverage
loans	type, principal, interestRate, tenureMonths, emiAmount, status	Pre-approved credit lines and active loans

rewards	rewardPoints, cashbackTotal, scratchCardsAvailable, activeVouchers	Gamified rewards and redeemable catalog items
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5. Detailed Screen Matrix (All 18 Stitch Screens)

- 1. **Screen 01 - Login:** Multi-method auth (Phone/Email) + "Continue as demo user" instant sign-in.
- 2. **Screen 02 - Home Dashboard:** Greeting, live wallet balance, 4 quick actions, 8 utility shortcuts, recent activity.
- 3. **Screen 03 - QR Scanner:** Live camera viewfinder overlay, flashlight toggle, gallery picker, manual UPI entry.
- 4. **Screen 04 - Contacts & Beneficiaries:** Search filter, frequent contacts carousel, full directory with tap-to-pay.
- 5. **Screen 05 - Pay & Numpad:** Custom responsive numpad, bank selector, note input, instant send action.
- 6. **Screen 06 - PayPause Sheet:** Visual friction sheet for large payments (>₹2,000) showing balance impact & 3s timer.
- 7. **Screen 07 - Payment Success:** Checkmark animation, transaction summary, sound haptics, share and receipt actions.
- 8. **Screen 08 - History & Spending Analysis:** FIChart interactive donut, monthly filter, outflow summary, category list.
- 9. **Screen 09 - Transaction Audit Sheet:** Detailed breakdown, reference ID, UTR number, repeat and dispute buttons.
- 10. **Screen 10 - Bills & Recharges Hub:** Grid of 10 utility categories, upcoming reminders, autopay status chips.
- 11. **Screen 11 - Autopay Setup Sheet:** Schedule selector (Instant, 3-day notice, Salary-linked) and monthly limit cap.
- 12. **Screen 12 - Credit Card Bill:** Card carousel, statement due dates, minimum vs custom pay, 1% fee calculation.
- 13. **Screen 13 - Wealth Dashboard:** Total portfolio net worth, returns graph, asset allocation bar, SIP shortcuts.
- 14. **Screen 14 - Digital Gold:** Live ticking 24K price, buy in rupees or grams, digital locker holding summary.
- 15. **Screen 15 - Insurance & Claims:** Health, auto, term life policy discovery cards and 4-stage live claim tracker.
- 16. **Screen 16 - Lending & EMI Calculator:** Instant eligibility slider (up to ₹2,50,000), tenure picker, live EMI math.
- 17. **Screen 17 - Rewards & Wallet:** Points balance, scratch cards with coin bursts, catalog voucher redemption.
- 18. **Screen 18 - Profile & Settings:** KYC badge, QR code generator modal, linked banks, security toggles, logout.